# 6-year statistical analysis of cloud occurrence and its physical properties using ACTRIS-CCRES remote sensing data

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**Abstract.**

Understanding climate change relies heavily on grasping the role of cloud radiative effect and its climate feedback. However, these factors pose the greatest uncertainty among atmospheric processes. This uncertainty stems partly from the limited global-scale cloud measurements. To address this gap, our study presents a six-year statistical analysis of cloud coverage and microphysics properties at the ACTRIS-CCRES AGORA observatory in Granada, Spain. Here, we utilized CLOUDNET post-processed data (e.g target classification, liquid water path, droplet effective radius and so on) of remote sensing instruments and microwave radiometer thermodynamic profiles. The analysis focuses on single layer cloud properties comparison for liquid, ice, mixed phase, liquid with rain, and mixed phase with rain clouds. Therefore, a distinct seasonal pattern in cloud occurrence was observed. During the summer (June-August), single layer clouds occurrence is less than 10%, and precipitation is absent. Conversely, cloud occurrence significantly increases during winter (December-February) and spring (March-May), where Ice and mixed phase clouds with rain are the predominant type. Additionally, a negligible presence of liquid clouds with rain were found. Seasonal trends in liquid water content (LWC) and droplet effective radius are unclear for mixed-phase and liquid clouds. Effective radius remains stable year-round, around 5 $\mu$m for liquid clouds below 3 km, while for mixed-phase clouds it can vary between 5 and 12 $\mu$m, mainly found at 1-2 km and 5-9 km. LWC shows higher values in cold periods and lower in summer, but with significant variability across both cloud types. Same seasonal patterns for ice water content (IWC) in ice and mixed-phase clouds were challenging to identify. Monthly profiles showed little variation, although values tended to rise in summer. Ice clouds typically have effective radius between 20-30 $\mu$m at 8-12 km altitude, peaking at 65 $\mu$m in summer. Mixed-phase clouds exhibit ice effective radius around 50 $\mu$m between 1-7 km altitude. The mean geometric thickness of ice and mixed-phase clouds throughout the seasons is 1.2 $\pm$ 1.3 km and 2.0 $\pm$ 1.6 km, respectively, contrasting sharply with the 0.2 $\pm$ 0.2 km observed in liquid clouds. The monthly variability in thickness is substantial enough to obscure any seasonal differences. Similarly, the analysis of cloud base height reveals mean values of 1.9 $\pm$ 2.0 km (excluding summer) for liquid clouds, 4.8 $\pm$ 2.2 km for mixed-phase clouds, and 7.0 $\pm$ 2.4 km for ice clouds. During summer a different vertical distribution of cloud base was found due to different mechanisms of cloud formation. Conclusively, a statistical analysis was conducted on single-layer cloud properties in the Southeast of Spain, highlighting the mainly cloud type associated with rainfall in the region.

# 1 Data and Methods

## 1.1 Data availability and station

An extensive dataset spanning the years 2018 to 2023 was utilized in this study. Specifically, high-level post-processed data from Cloudnet (**?**), including levels 1c and 2 (categorize, classification, and microphysical files), along with microwave radiometer humidity and temperature profiles, were employed. Figure 3 illustrates the monthly data availability for the 6-year Cloudnet post-processed dataset at the AGORA ACTRIS-CCRES station (CCRES stands for Centre for Cloud Remote Sensing, and AGORA for Andalusian Global ObseRvatory of the Atmosphere), situated within a broad intramontane depression in the foothills of Sierra Nevada, Spain. This figure also highlighted the available percentage of data per year, which were obtained by a simultaneous availability of RPG-HATPRO radiometer, CHM15k ceilometer, RPG-FMCW 94 GHz cloud Doppler radar and European model ECMWF (acronym for European Centre for Medium-Range Weather Forecasts) data.

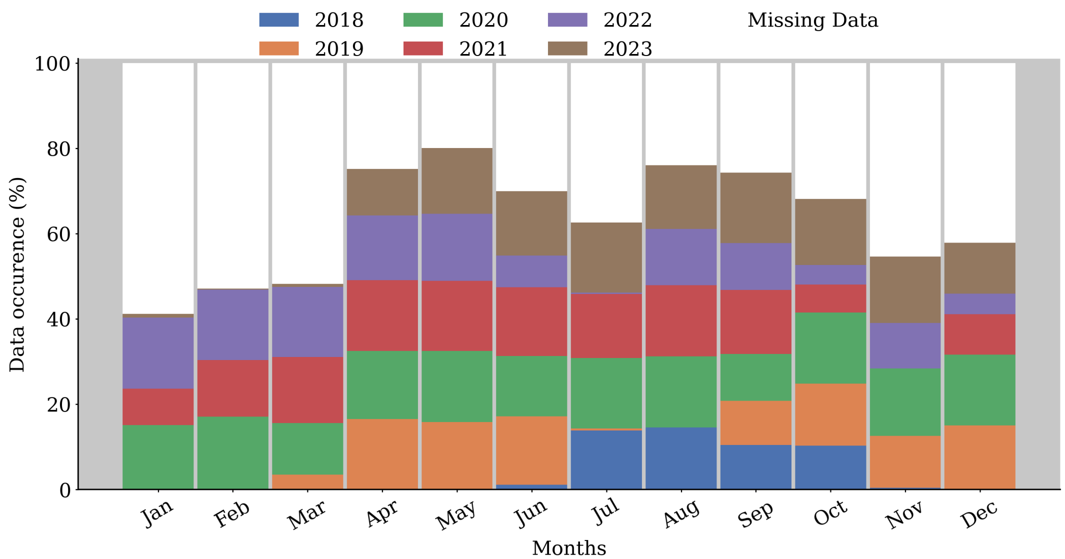


**Figure 1.** ...

The dataset shows that data collection efforts improved over time, with 2020 exhibiting the highest and most consistent data availability throughout the year. In contrast, 2018 had no data for the early months, indicating that data collection commenced mid-year. The gray bars denote periods with missing data due to instrument maintenance and technical issues. Despite these challenges, these three instruments have been continuously operated since 2018.

It is worthy to note that this dataset represents a robust database for cloud statistics, providing comprehensive data for cloud analysis over an extended period. Notably, it stands as the only cloud remote sensing dataset available in Spain, making it an invaluable resource for regional climate studies. The ACTRIS-CCRES Granada site is uniquely positioned as one of the few locations dedicated to cloud studies in a warm region that is frequently affected by severe weather conditions **?**. This dataset contributes significantly to filling the gap in cloud data for this climate variability hotspot, enhancing our understanding of cloud dynamics and aiding in the improvement of predictive models and climate studies in these critical areas.

### 1.2 Instrument and products

Overall, instrument datasets were processed by Cloudnetpy (**?**), which sampled data on a common grid with a time resolution of 30,s and height bins based on the cloud radar range resolution, which varies according to the radar chirp table configuration. This table provides the range resolution at different levels, the maximum unambiguous velocity, and the maximum range reached by the radar. These three variables are interdependent and can be strategically adjusted to meet specific objectives. In our case, this table was modified several times throughout the analyzed period. In addition, range resolution of temperature and humidity profiles given by the microwave radiometer also changed with time. Consequently, to ensure consistency in the vertical resolution of the data, a regridding process was implemented to facilitate monthly statistical analysis when needed. The regridding involved interpolating measurements onto a height grid with a 30 m resolution and was applied to every inter-annual profiling. Additionally, CloudnetPy performs corrections for reflectivity gas and liquid attenuation, error estimation, and quality flag assignment. Specifically, liquid attenuation corrections on radar reflectivity are crucial during raining events, as rain droplets can attenuate a significant fraction of the radar signal **?**. Table 1 provides a summary of the cited instruments, detailing their operational frequencies or wavelengths, the types of measurements they perform, their products, their measurement ranges, data collection intervals, and relevant references for a detailed description. The unique variables not processed by Cloudnetpy were humidity ($q$) and temperature (T) profiles, given by the microwave radiometer. Also, specially important high level data such as liquid water content (IWC), ice water content (IWC), droplet effective radius and ice effective radius is described in table XXX. For each product in this table, the necessary input data and methods used for their computation are specified alongside their associated uncertainty percentages and the studies from which these products are sourced.

**Table 1.** This is a table with some data.

| Instrument | Bands / wavelenghts | Measured | Products | Range | Time | References |
|---|---|---|---|---|---|---|
| 94 GHz Cloud Doppler Radar | 94 GHz (W-band) | Doppler velocity spectrum (DVS) | $\nu_D$ , Z, $S_w$ | [13,51] m[a] | [1.1, 3.6] s[b] | Nils Küchler et al. |
| MWR RPG | 22-31 GHz (K-bad) | Brightness | q and T | [10, 300] m[c] | [2, 2.5] min | Francisco Navas-Guzmán et al. |
| HATPRO | 51-58 GHz (V-band) | Temperature | LWP | - | 2 s | |
| CHM15k Nimbus Ceilometer | 1064 nm | | $\beta$ | 15 m | 15 s | A. Cazorla et al. |

| Cloudnet products | Input role | $\sigma_{re}/r_e$ | References |
|---|---|---|---|
| Liquid water content (LWC) | Z and $\beta$: cloub base and top identification<br>$T_{ECMWF}$, $P_{ECMWF}$: adiabatic LWC<br>MWR or CDR LWP to scale LWC | 15% | Frisch et al. (1998) |
| Ice water content (IWC) | Z and $T_{ECMWF}$ profiles | 30-40% | Robin J. Hogan et al.<br>Julien Delanoë et al. |
| Droplet effetive radius ($r_{liq}$) | Z, and N and $\sigma$ of lognormal PSD | 15% | S. Frisch et al. |
| Ice effective radius ($r_{ice}$) | Z and $T_{ECMWF}$ | 50% | Hannes Griesche et al. |

### 1.2.1 Cloudnet particle classification scheme

Target classification product is the core of these study, it assigns an integer ranging from 0 to 10 at each pixel in the Cloudnet processed grid. Each number are related to atmospheric elements in the following order: clear sky, droplets, drizzle or rain, drizzle & droplets, ice, ice & droplets, melting ice, melting & droplets, aerosol, insects, aerosol & insect. These targets were determined through radar and lidar variables in combination with thermodynamic profiles from the ECMWF model. In particular, wet bulb temperature ($T_w$) profiles is used to establish warm and cold regions to classify liquid and ice hydrometeors. Further, liquid droplets boundaries were resolved with lidar attenuated backscatter coefficient ($\beta$), and radar reflectivity. The first is sensible to small liquid droplets present in cloud base, as the second are able to penetrate the cloud and detect cloud tops. In addition, liquid layers respect a threshold of T < -40 °C, since it is not possible to keep water in liquid phase bellow that temperature. Hence, liquid droplets assigned as falling were classified as drizzle or rain droplets, where falling is attributed through radar echos analysis once particles boundary were already determined. Therefore, falling cold particle were just classified as ice, without distinguishing ice from precipitating ice. Next, melting layers were identified at the height of maximum wet bulb temperature divergence. Pixels with divergence surpassing 0.0075 $s^{-1}$ within 150 m of height deviation were also classified as "melting". Furthermore, more than one class of particle can be settled in the same pixel, allowing the identification of mixed phase hydrometeors, like ice coexisting with supercooled liquid droplets and melting ice particles coexisting with cloud liquid droplets. A more detailed description of particle apportionment of Cloudnet algorithm can be found on **?**.

## 1.3 Cloud classification algorithms

### 1.3.1 Old classification scheme

The assortment algoritm classifies groups of hydrometeors and cloud layers. In terms of hydrometeors, this algorithm generate masks of hydrometeor occurrence for 3 groups: liquid, ice and "total", where liquid class encompass all possible liquid pixels, which are "droplets", "drizzle or rain" and "drizzle & droplets". Ice encompass only "ice" pixels, and total is all possible

[a] This range vary with the height, and depends on the chirp table configuration

[b] It also depends on the chirp type

[c] Humidity and temperature profiles has a range varying from 10 to 150 m in the first 4.4 km and from 50 to 300m between 4.4 and 10 km

hydrometeors, which are "droplets", "drizzle or rain", "drizzle & droplets", "ice", "ice & droplets", "melting ice" and "melting & droplets".

Subsequently, cloud layers in atmospheric column were classified trough pixel sequence inquiry. These pixels sequences were divided in the following 5 groups according to **?**: liquid, ice, mixed phase, liquid with precipitation and mixed phase with precipitation. Specifically, liquid clouds were assigned as any combination of "droplets" and "drizzle & droplets" pixels. Ice clouds were assigned when only "ice" pixel sequence was encountered. As well as, mixed phase clouds were assigned with any combination of "droplets", "ice", "ice & droplets", "melting ice", "melting & droplets", "drizzle and droplets" pixels. Precipitation cases were classified whether at least one pixel of "drizzle or rain" is found below liquid and mixed phase clouds. Different from previous studies, a threshold of 20 bins (256-1022 m) bellow cloud was set to reject precipitation events, assuming that rain droplets are not falling from the actual cloud layer. In all cases, for being considered a cloud layer, a minimum sequence of 3 cloud pixels (38-153 m) (**?**) must be satisfied, in addition to a minimum sequence of 5 pixels (64-255 m) of "Clear sky", "Aerosol", "Insect" and "Aerosol & insect" to separate multi-layers. Therefore, free hydrometeor layers in the neighbourhood of the cloud allowed the identification of cloud base and top. Nevertheless, thresholds here were based on **??**, but theses values are not currently well defined, their determination has been guided by empirical experience.

### 1.3.2 A novel classification scheme

To enhance the classification of hydrometeors using the Cloudnet algorithm, we first identified hydrometeor pixels and created a cloud mask with zeros and ones representing their positions. We then applied a dilation method to expand the ones in the mask, thereby connecting the closest groups of hydrometeors, which are likely to belong to the same cloud. Next, we identified groups of connected ones to classify them into distinct clouds. Each group was then iterated over, followed by a contraction to the original resolution to preserve the indices of the dilated pixels. Rain pixels were counted and subsequently removed from each group, as the number of raining pixels would be used later to classify the clouds as precipitable or non-precipitable. However, before anything else, to be classified as a cloud, each group needed to contain at least 100 pixels; otherwise, it was deemed noise. Within each cloud group, we computed the percentage of each hydrometeor type, excluding rain pixels, to follow a specific classification scheme, also represented in the flow chart XXX: if a cloud comprised more than 70% Droplets and Drizzle & Droplets, it was classified as a liquid cloud. If it contained at least 10 connected rain pixels, it was further classified as a precipitable liquid cloud; otherwise, it remained classified as a non-precipitable liquid cloud.

Similarly, if a cloud contained more than 90% ice or less than 10% Droplets plus Ice & Droplets, it was classified as an ice cloud. With at least 10 connected rain pixels, it was categorized as a precipitable ice cloud; otherwise, it remained a non-precipitable ice cloud. To avoid an underestimation of cloud base height due to drizzle, which only appears below the melting layer, we removed these pixels. This was observed due to an extensive analysis of individual precipitable ice clouds. Thus, if a cloud did not meet the conditions for liquid or ice clouds, it was classified as a mixed-phase cloud. If it contained at least 10 connected rain pixels, it was classified as a precipitable mixed-phase cloud; otherwise, it was classified as a non-precipitable mixed-phase cloud.

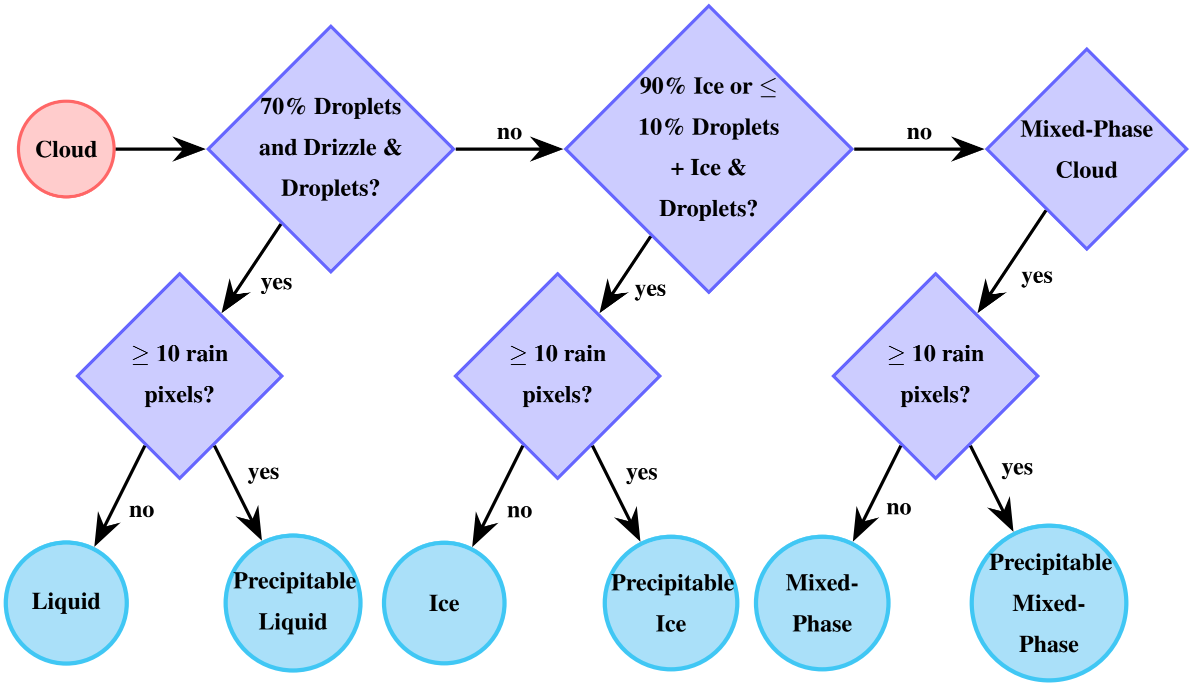


**Figure 2.** Flowchart for cloud classification based on hydrometeor type percentage and the presence of connected rain pixels.

In addition to the primary classification scheme, each cloud was further classified over time to distinguish between single and multi-layer clouds. Initially, we identified the time intervals where cloud groups overlapped. Overlapping times were classified as multi-layer, whereas non-overlapping times were classified as single layer. We then analyzed individual cloud layers, acknowledging that a single cloud could exhibit multiple layers within a given time interval. If a cloud had a second layer separated by only one pixel at the i-th time step, that time step was classified as multi-layer. Otherwise, clouds were classified as single layer, and their physical properties were calculated as follows: the cloud base and top at each i-th time step were determined by the first and last height pixels within the cloud layer, respectively. Cloud thickness was computed as the difference between the cloud top and base.

A final verification was conducted for ice-precipitable clouds due to frequent misclassifications observed in several cases, exemplified by figure XXX. This figure illustrates a thinner and lower precipitable ice cloud present between 12:00 and 16:00, undetected by lidar backscattering and radar reflectivity. Analysis of numerous similar cases indicated that these clouds, which are likely misclassifications, erroneously doubled the observed frequency of cloud occurrence in May. To prevent this misclassification, we filtered all precipitable clouds with a mean cloud base below 4 km above mean sea level and an average cloud thickness below 700 meters, as typical precipitable ice clouds are usually much thicker. These clouds were reclassified as noise. Finally, time intervals with no clouds were classified as clear sky. It is important to note that noise can coexist with clear sky, but single layer, clear sky, and multi-layer classifications are mutually exclusive.

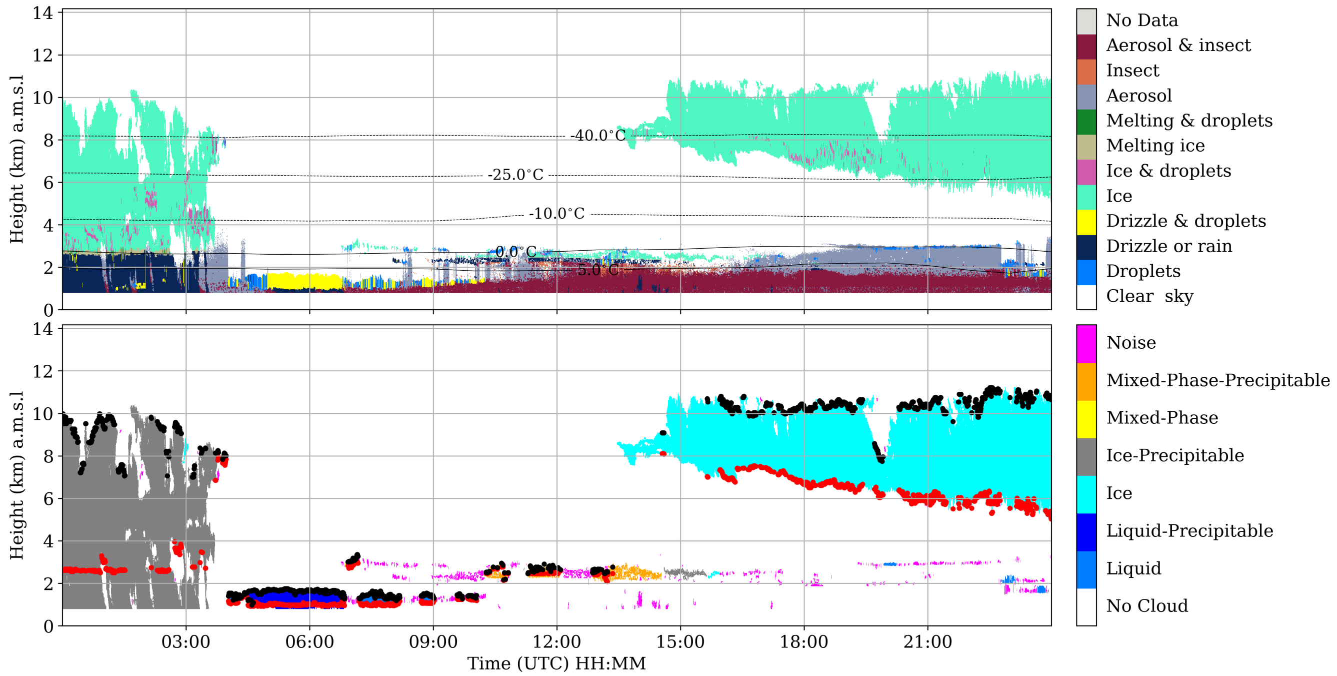


**Figure 3.** Comparison of cloud classification methods. The left panel illustrates the old method used in recent studies, where cloud classification is done by time. The right panel shows the new method, which preserves cloud homogeneity and provides a more realistic classification.

Figure 1 shows a comparison between the old and new methods of cloud classification. The left panel illustrates the old method, which was employed in some recent studies. This method classifies clouds based on time, leading to the fragmentation of a single cloud into different types. The right panel presents the new method proposed in this study, which preserves the homogeneity of clouds, providing a more realistic classification.

The conclusion from this comparison is that the traditional method of cloud classification can generate high correlations between clouds of similar composition, such as ice and mixed-phase clouds, which both have high concentrations of ice. This results in a strong correlation in their physical properties, as will be discussed in section XXX through the daily ice water content analysis. This strong correlation is expected since the old method allows the classification of multiple cloud types within the same cloud. The new method, by maintaining the homogeneity of clouds, avoids this issue and provides a more accurate representation of cloud properties. This comprehensive approach ensures more accurate cloud classification and property calculation, reducing errors and enhancing our understanding of cloud dynamics.

*Acknowledgements.* TEXT

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